Ofqual QAN	**600/3055/0**
Quartz ID	**1158 / 1159 / 1161**
Qualification	**Level 5 Diploma in Computing / Level 5 Diploma in Computing with Business Management / Level 5 Diploma in Computing with Data Science**

Assessment	**Introduction to Data Science and Big Data**
Assessment Type	**Examination**
Date	**Tuesday 4th March 2025**
Time	**09.00am – 12.00pm**

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) must be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC education materials should be the first authority.

This marking scheme has been prepared as a guide only to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Marker's comments:		
Moderator's comments:		
Mark:	**Moderated mark:**	**Final mark:**
Penalties applied for academic malpractice:		

Answer ALL questions

Marks

Question 1

a) Define big data **and** explain how it differs from traditional databases. Discuss the key characteristics of big data. **5**

Mark scheme:

Big data refers to extremely large and complex datasets that are difficult to process using traditional data management tools (1). Unlike traditional databases, which handle structured data within predefined schemas, big data systems manage dynamic, unstructured and complex datasets (1). It is characterised by the 5Vs: Volume, representing the massive size of the data that exceeds traditional database capabilities (1); Velocity, indicating the high speed at which data is generated and processed, often in real time (1); Variety, which encompasses the diverse formats of data, including structured, unstructured and semi-structured forms (1); Veracity, highlighting the importance of data accuracy and reliability despite inconsistencies (1); and Value, emphasizing the potential insights and business benefits derived from analysing big data (1).

Allocate maximum 3 marks for characteristic (3Vs out of 5Vs). Any suitable alternative explanation or valid characteristic can also be credited, with a maximum of 5 marks awarded.

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		Marks
b)	Provide an example of how big data is being used in healthcare or finance.	**5**

Mark scheme:

In healthcare, big data is used to analyse vast amounts of patient data from electronic health records, wearable devices and medical imaging (1). This enables predictive analytics to identify patterns that help in diagnosing diseases early (1), personalising treatments based on patient history (1) and predicting potential outbreaks of diseases (1). For example, big data can analyse patient trends to foresee hospital admissions, allowing better resource allocation and improving overall patient care (1). This use of big data enhances both efficiency and the quality of healthcare services (1).

OR

In finance, big data is used to detect fraudulent activities by analysing large volumes of transaction data in real-time (1). It helps identify unusual patterns or anomalies, allowing banks and financial institutions to flag and prevent fraudulent transactions quickly (1). Big data is also used in risk management, where predictive models assess market risks by analysing historical data and current market trends (1). For instance, credit scoring systems use big data to evaluate a customer's financial behaviour and predict their creditworthiness more accurately (1). This improves decision-making and enhances security within the financial industry (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks.

Total 10 Marks

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Marks

Question 2

a) Explain the steps involved in the data science lifecycle. **5**

Mark scheme:

The data science lifecycle involves the following steps:
Understanding Business and Data: Aligning the project goals with data requirements (1).
Data Collection: Gathering relevant data from various sources (1).
Data Pre-processing: Cleaning and preparing the data for analysis (1).
Data Analysis: Exploring patterns and insights through initial analysis (1).
Build Model: Training the model using appropriate algorithms and training data (1).
Model Evaluation: Testing the model's performance with test data (1).
Model Deployment: Deploying the model to generate predictions or insights (1).
Knowledge and Action: Using insights from the model for informed decision-making (1).
This process ensures a structured approach to data-driven solutions.

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Discuss the importance of pre-processing in the data science lifecycle. Provide examples of how errors in the steps in a) can impact project outcomes. **5**

Mark scheme:

Data pre-processing is a crucial step in the data science lifecycle, as it ensures that the data is clean, consistent and ready for analysis (1). This involves handling missing data, correcting errors, removing duplicates and normalising or scaling data to prepare it for modeling (1). For example, failing to address missing values in a dataset can lead to biased models, as certain groups might be underrepresented (1). Similarly, if data outliers are not addressed, they can distort model predictions and reduce accuracy (1). Pre-processing also ensures better model performance and interpretability, as well-prepared data allows machine learning algorithms to focus on meaningful patterns rather than noise (1).

Any suitable alternative allowed such as storage issue, data inconsistency etc, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

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Marks

Question 3

a) What are categorical and continuous data types? Provide examples of each. **5**

Mark scheme:

Categorical data refers to data that can be divided into distinct categories or groups (1).
Examples include gender (male, female), colours (red, blue, green), or types of cars (sedan, SUV) (1). This type of data often represents labels or qualities rather than numerical values (1).

Continuous data consists of data that can take any numerical value within a range (1).
Examples include height, weight, temperature or time (1). Continuous data can be measured with greater precision and often represents quantities on a scale (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Explain the concept of Principal Component Analysis (PCA) and its importance in data analysis. Provide an example of when PCA might be used. **5**

Mark scheme:

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms high-dimensional data into a smaller number of components while retaining most of the variance in the dataset (1). It works by identifying principal components, which are linear combinations of the original features, ordered by the amount of variance they capture (1). PCA is important in data analysis because it reduces complexity, mitigates overfitting in machine learning models and improves visualisation of high-dimensional datasets (1). For example, PCA can be used in image compression to reduce the number of pixels needed to represent an image without significant loss of quality (1). Another use case is in genomic data, where thousands of gene expression values are reduced to a smaller number of components for analysis (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

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Marks

Question 4

a) Describe how missing values occur in a dataset **and** why they are problematic. **5**

Mark scheme:

Missing values in a dataset occur when no data is stored for a variable in one or more observations (1). These can arise from data collection errors, system issues or incomplete responses (1). Missing values are problematic because they can lead to biased results, distort analysis and reduce the accuracy of machine learning models (1). Additionally, many algorithms cannot handle missing data directly, which can prevent models from being trained properly (1). Therefore, addressing missing values is crucial for maintaining the quality and reliability of the analysis (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Discuss **TWO (2)** methods for handling missing values in data analysis. You should identify advantages and disadvantages of each. **5**

Mark scheme:

Deletion: This involves removing rows or columns with missing data (1). It is suitable when the missing values are minimal and not critical to the analysis (1). However, if too much data is deleted, it can reduce the dataset's representativeness (1).

Imputation: This method replaces missing values with substituted data, such as the mean, median, or mode for numerical data (1). It preserves the dataset size and avoids losing valuable information, but if not done carefully, it can introduce bias (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

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Marks

Question 5

a) Explain in detail the difference between exploratory data analysis **and** confirmatory data analysis. **6**

Mark scheme:

Exploratory Data Analysis (EDA) is used to investigate and discover patterns, trends and relationships within a dataset without specific hypotheses (1). It helps in understanding data structure, identifying anomalies and generating insights (1). EDA is open-ended and often visual, using tools like scatter plots, histograms and box plots to explore the data (1).

Confirmatory Data Analysis (CDA), on the other hand, is hypothesis-driven (1). It tests predefined hypotheses and assesses whether the data supports or rejects them using statistical methods such as t-tests or regression analysis (1). CDA focuses on validation and proving or disproving specific assumptions, making it more structured and formal than EDA (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 6 marks

b) Provide an example with explanation where **both** approaches in a) would be applied in a data science project. **4**

Mark scheme:

In a customer segmentation project for a retail business, Exploratory Data Analysis (EDA) might be used initially to explore customer purchasing behaviours, identifying patterns or clusters of customers based on spending habits or preferences (1). For example, EDA could reveal potential segments like high spenders, budget shoppers and occasional buyers (1).

After identifying these patterns, Confirmatory Data Analysis (CDA) could then be applied to test specific hypotheses, such as whether high spenders respond better to a certain marketing campaign (1). CDA would use statistical methods to confirm or refute the effectiveness of the campaign across the identified segments (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks

Total 10 Marks

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Marks

Question 6

a) Explain in detail what is meant by a classification model in supervised learning. **5**

Mark scheme:

A classification model in supervised learning is an algorithm that categorises or labels data into predefined classes or categories (1) based on input features (1). The model is trained using labelled data, where the correct output (class) is provided for each input (1). The goal is for the model to learn the relationship between the input features and the corresponding class labels, so it can accurately predict the class of new, unseen data (1). Examples of classification tasks include spam detection in emails or classifying images of animals into categories like "cat" or "dog" (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Discuss the importance of model evaluation in ensuring model accuracy. **5**

Mark scheme:

Model evaluation is crucial for ensuring the accuracy and reliability of a machine learning model (1). It helps assess how well the model performs on unseen data, (1) ensuring it generalises beyond the training set (1). By using evaluation metrics like accuracy, precision, recall and F1-score, data scientists can determine if the model is making accurate predictions (1). Evaluation also helps detect issues like overfitting, where a model performs well on training data but poorly on new data (1). Ultimately, model evaluation ensures that the model is effective for real-world applications and decision-making (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

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Marks

Question 7

a) Discuss the challenges of visualising multi-dimensional data **and** explain which types of plots or charts can help to overcome these challenges. **6**

Mark scheme:

Visualising multi-dimensional data is challenging because traditional 2D or 3D plots cannot easily represent the relationships between many variables (1). As the number of dimensions increases, it becomes harder to detect patterns, trends or correlations (1). One way to overcome this is by using scatter plot matrices, which display pairwise relationships between variables in a grid of scatter plots (1). Another approach is parallel coordinates plots, where each axis represents a dimension, allowing patterns to be visualised across multiple variables (1). Additionally, heatmaps can be used to represent multi-dimensional data by showing the correlation between different variables using colour intensity (1). These techniques help to simplify complex data, making it easier to interpret (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 6 marks

b) Discuss **TWO (2)** examples where advanced visualisation techniques for multi-dimensional data are essential for decision-making **and** explain why these techniques are effective in those scenarios. **4**

Mark scheme:

One example is in financial portfolio management, where advanced visualisation techniques like parallel coordinates plots are essential for comparing multiple investment factors, such as risk, return and asset type, across various portfolios (1). This allows portfolio managers to visually assess and optimise their asset allocation based on different criteria simultaneously (1).

Another example is in healthcare, where heatmaps can be used to show correlations between various patient metrics such as age, blood pressure and cholesterol levels (1). This helps doctors and healthcare professionals identify patterns and risk factors across multiple variables, leading to better diagnosis and treatment decisions (1). These advanced visualisation techniques simplify complex, multi-dimensional data, making insights easier to interpret for informed decision-making (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 4 marks

Total 10 Marks

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Marks

Question 8

a) Discuss the importance of evaluation metrics in model scoring. Identify **and** explain **FOUR (4)** evaluation metrics used to assess model performance **and** provide examples of when each would be most appropriate to use. **10**

Mark scheme:

Evaluation metrics are essential in model scoring as they provide a quantitative assessment of how well a model performs in making accurate predictions (1). These metrics help to determine whether a model is suitable for the task at hand and ensure that it generalises well to new data (1). They also allow data scientists to compare different models and select the best one for a particular problem (1). Without proper evaluation, a model may perform poorly in real-world scenarios, leading to inaccurate predictions and ineffective decision-making (1).

Four common evaluation metrics include:
Accuracy: Measures the proportion of correct predictions out of the total predictions (1). Accuracy is most appropriate when the class distribution is balanced, such as in spam detection, where there are roughly equal numbers of spam and non-spam emails (1).
Precision: Refers to the proportion of true positive predictions out of all positive predictions (1). Precision is crucial when false positives are costly, such as in medical diagnoses, where wrongly predicting a disease could lead to unnecessary treatment (1).
Recall (Sensitivity): Measures the proportion of true positives out of all actual positives (1). It is important when missing positive cases is costly, such as in fraud detection, where it is essential to identify as many fraudulent cases as possible (1).
F1 Score: The harmonic mean of precision and recall, balancing the two metrics (1). F1 score is useful when there is an uneven class distribution, as it accounts for both false positives and false negatives, such as in imbalanced datasets like rare disease detection (1).

Each metric offers a different perspective on model performance, allowing for a more complete evaluation (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 10 marks

Total 10 Marks

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Marks

Question 9

a) Explain the role of advanced BI (Business Intelligence) tools in predictive analytics **and discuss** how they contribute to proactive decision-making in industries like finance or healthcare. **10**

Mark scheme:

Advanced Business Intelligence (BI) tools play a crucial role in predictive analytics by enabling organisations to analyse historical data (1), identify patterns (1) and forecast future outcomes (1). These tools use techniques like machine learning (1) and statistical modelling (1) to make accurate predictions, helping businesses anticipate future trends and challenges (1).

In finance, BI tools are used for predicting stock prices (1), assessing credit risk (1) and detecting fraud (1). By analysing vast amounts of transaction data in real-time (1), financial institutions can make proactive decisions to mitigate risk or capitalise on investment opportunities (1). For example, predictive analytics can forecast market trends, allowing investors to adjust their portfolios in advance (1).

In healthcare, BI tools help predict patient outcomes (1), disease outbreaks (1) and resource needs (1). Hospitals can use these insights to optimise treatment plans (1), manage patient loads (1) and allocate resources more efficiently (1). For instance, predictive models can help foresee patient admission rates, enabling better staffing and resource management (1).

Overall, advanced BI tools enhance proactive decision-making by providing actionable insights, enabling businesses to address challenges before they arise and take advantage of future opportunities (1). This leads to improved operational efficiency, reduced risk and better strategic planning in industries like finance and healthcare (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 10 marks

Total 10 Marks

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Marks

Question 10

a) Discuss **ONE (1)** ethical issue related to data aggregation in big data projects. **5**

Mark scheme:

One ethical issue in data aggregation is the loss of individual privacy (1). When data from various sources is combined, even anonymised data can unintentionally reveal personal information (1). For example, aggregating social media and purchase data can expose sensitive details about individuals (1). This raises concerns about consent and the responsible use of personal data (1), making it vital for organisations to ensure transparency and protect privacy during aggregation (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

b) Explain how organisations ensure accountability **and** transparency in the use of big data. **5**

Mark scheme:

Organisations can ensure accountability and transparency by implementing clear data governance policies (1), conducting regular audits (1) and publishing transparency reports on data usage (1). Using ethical frameworks and ensuring open communication with stakeholders builds trust (1), while employee training reinforces accountability in data handling (1).

Any suitable alternative allowed, 1 mark for each correct point to a maximum of 5 marks

Total 10 Marks

End of paper

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Learning Outcomes matrix

Questions	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO 1	Yes
2	LO 1, 2	Yes
3	LO 2, 3	Yes
4	LO 2, 3	Yes
5	LO 2, 4	Yes
6	LO 1, 2, 4	Yes
7	LO 4	Yes
8	LO 1, 5	Yes
9	LO 1,4, 5	Yes
10	LO 1, 5, 6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Demonstrate a systematic and thorough understanding of the concept of Data Science and Big Data	Exhibits minimal grasp of foundational concepts in Data Science and Big Data. Fails to showcase an ability to identify, adapt, or utilize appropriate skills, methods, and procedures effectively in addressing data-related challenges.	Displays an insufficient understanding of essential concepts in Data Science and Big Data. Struggles to articulate and apply appropriate skills, methods, and procedures adequately, indicating a need for further development.	Achieves a satisfactory level of understanding in the fundamental concepts of Data Science and Big Data. Demonstrates the ability to identify, adapt, and apply foundational skills, methods, and procedures to address standard data challenges.	Attains a very good understanding of advanced concepts in Data Science and Big Data. Exhibits a proficient ability to identify, adapt, and effectively use a range of skills, methods, and procedures, surpassing basic requirements.	Attains an outstanding level of comprehension in intricate concepts of Data Science and Big Data. Demonstrates an exceptional ability to identify, adapt, and apply a comprehensive set of skills, methods, and procedures, yielding well-supported, highly nuanced solutions that showcase depth and mastery.
Demonstrate proficiency in data collection, design, and modelling for data processing	Struggles to demonstrate basic proficiency in data collection, design, and modelling for data processing. Exhibits limited understanding of fundamental concepts and is unable to apply them effectively.	Displays some proficiency in data collection, design, and modelling for data processing but falls short of the expected standard. Shows inconsistencies or weaknesses in applying relevant skills, methods, and procedures.	Demonstrates a satisfactory level of proficiency in data collection, design, and modelling for data processing. Applies foundational skills, methods, and procedures effectively, achieving competence in standard scenarios.	Attains a very good level of proficiency in data collection, design, and modelling for data processing. Demonstrates a well-developed ability to apply a range of skills, methods, and procedures effectively, surpassing	Achieves an exceptional level of proficiency in data collection, design, and modelling for data processing. Displays mastery in applying a comprehensive set of advanced skills, methods, and procedures, showcasing depth and excellence in addressing complex data processing challenges.

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				basic requirements.	
Select appropriate techniques and tools for data pre-processing.	Struggles to select or apply basic techniques and tools for data pre-processing	Shows some ability to select techniques and tools but falls short of the expected standard.	Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.	Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.	Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.
Understand the analytical techniques and software tools to effectively generate insights from data.	Struggles to select or apply basic techniques and tools for data pre-processing	Shows some ability to select techniques and tools but falls short of the expected standard.	Demonstrates a satisfactory ability to select appropriate techniques and tools for data pre-processing.	Attains a very good ability to select and apply a range of techniques and tools for data pre-processing.	Achieves an exceptional ability to select and apply advanced techniques and tools for data pre-processing.
Understand the suitability of different visualisation methods for different types of Data	Struggles to understand the basic suitability of visualization methods for different types of data.	Displays some understanding of the suitability of visualization methods but falls short of the expected standard.	Demonstrates a satisfactory understanding of the suitability of visualization methods for different types of data.	Attains a very good understanding of a range of visualization methods and their suitability for different data types.	Achieves an exceptional understanding of advanced visualization methods and their nuanced suitability for different data types.
Demonstrate a critical understanding of Data Science Ethics	Struggles to demonstrate a basic understanding of data science ethics.	Displays some understanding of data science ethics but falls short of a critical analysis.	Demonstrates a satisfactory understanding of data science ethics.	Attains a very good understanding of a range of ethical considerations in data science.	Achieves an exceptional understanding of advanced ethical considerations in data science.

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